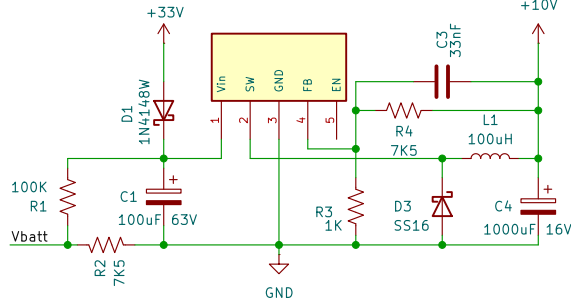
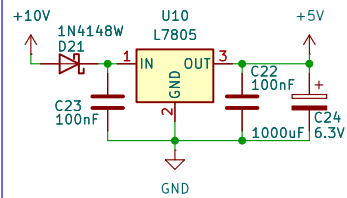


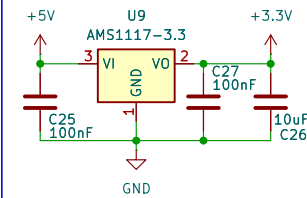
DC-DC Step down 33 -> 10V 150 mA



5V linear power supply

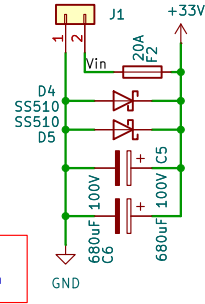


3.3V linear power supply



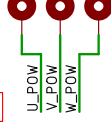
MCU

Power connector



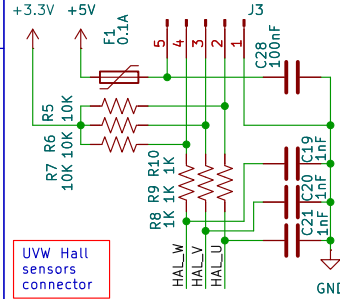
Reverse polarity protection

Motor connector



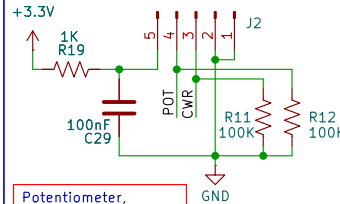
Motor connectors

Hall sensors connector



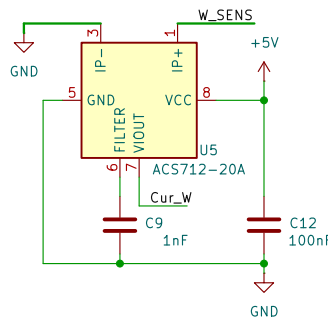
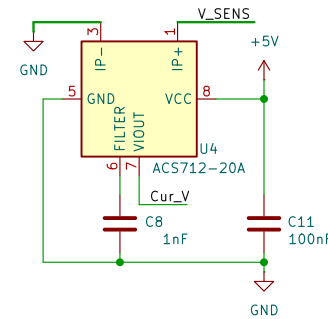
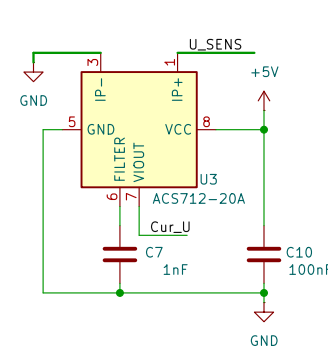
UVW Hall sensors connector

Torque control connector

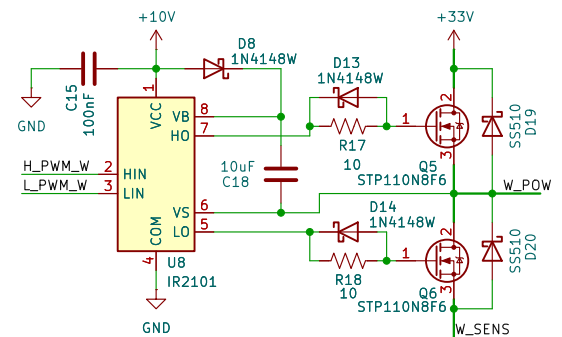
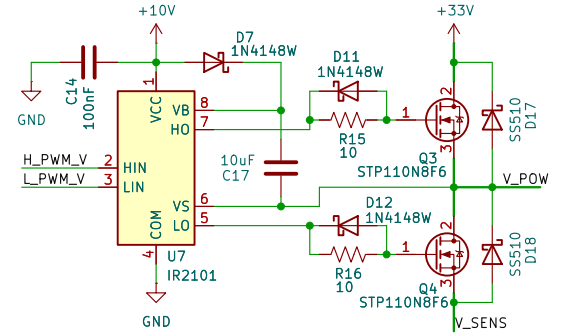
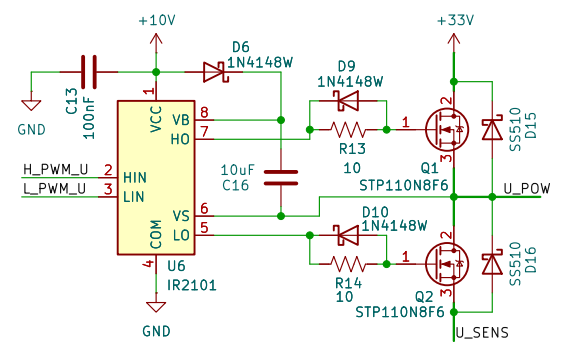


Potentiometer, enable pin and type rotate set connector

Current sensors(isolated)



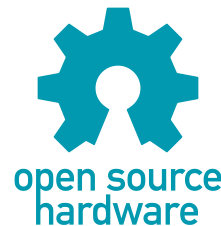
MOSSFET and drivers(6PWM)



The IR2101(HIN/LIN) has internal pull-down.

3-Phase STM32 BLDC motor driver:

- Battery: 8S
- VCC: 24.0V to 33.6V
- I(max): 20A
- Current measurements per phase



By Techn0man1ac

Sheet: /

File: Inverter.kicad_sch

Title: Simple STM32 BLDC motor controller

Size: A4

Date: 2026-06-14

KiCad E.D.A. 10.0.3

Rev: V0.2

Id: 1/1